

WASP-12b

R-1982

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_151030 · created 2026-08-07T00:56:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457324.84055 +/- 0.00022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.043
Transit depth (Rp/Rs)^2	0.00045 +/- 0.018 [%]
Orbital Inclination (inc)	79.9 +/- 2.89
Airmass coefficient 1 (a1)	5.27 +/- 4.12
Airmass coefficient 2 (a2)	-1.58 +/- 0.17
Scatter in the residuals of the lightcurve fit is	50.13 %
Best Comparison Star	#1 - [624, 481]
Optimal Aperture	4.69
Optimal Annulus	12.43
Transit Duration (day)	0.0984 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.002 ± 0.043 vs 0.1178 (deviation 1.01σ)
Duration fit vs published	0.0984 d vs 0.125 d (deviation 1.02σ)
Mid-time O–C	-1571.6 min
Depth SNR	0.0
Residual scatter	50.13 %

Light curve

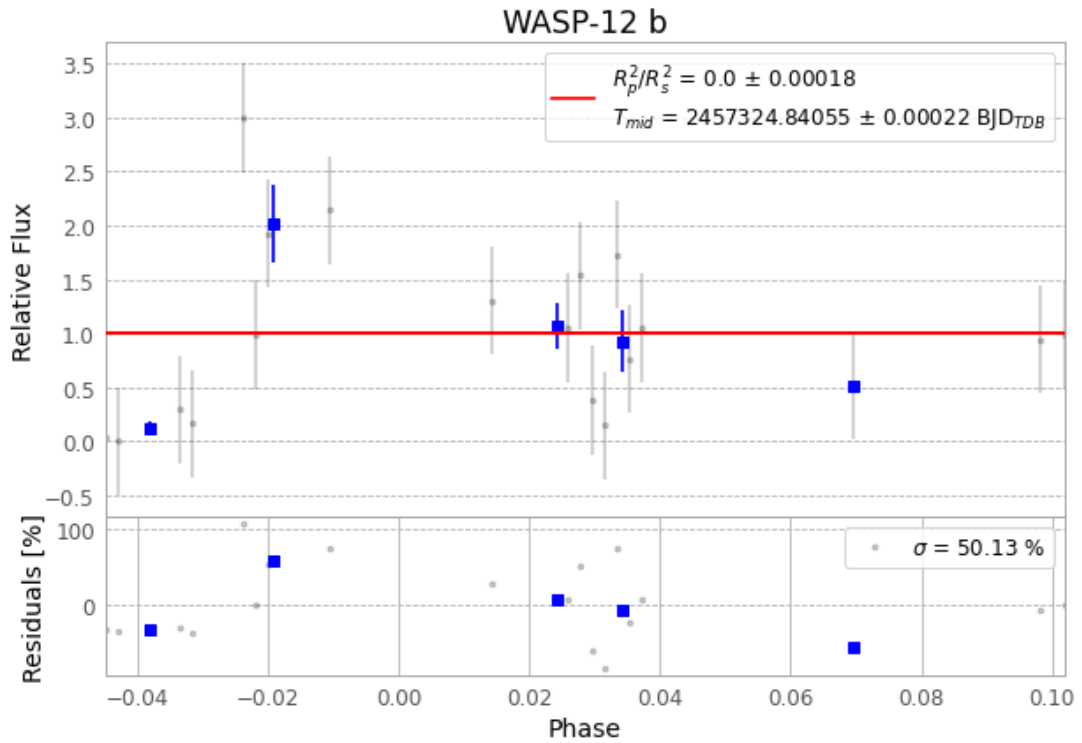


Figure 1 — FinalLightCurve_WASP-12 b_2015-10-28.png (SHA-256 5e67c76952ffcf8d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [279, 317], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 31a1fec26af02cfe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

84 FITS frames (55 MB), WASP-12151029063912.FITS → WASP-12151029104812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151030.log (1ecc198c5d76...), FinalLightCurve_WASP-12 b_2015-10-28.png (5e67c76952ff...), FinalLightCurve_WASP-12 b_2015-10-28.csv (f53a287895c6...)

Compute resources

Wall time 122.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:56Z.